# Directed Cyclic Networks
# Research Contract and Proof Compendium

P02 audit closure: 27 September 2026

**Current mathematical status: P01 is VERIFIED / AUDITED; P02 is VERIFIED / AUDITED; P03–P18 are TODO.** The independent audit `audits/A01_wellposedness_and_absorbing_box.tex` records CORE VERIFIED and EXTENDED MINOR REVISION. Its sole required correction, E01, is applied: the archived radius is described as the larger real root and as nonnegative. The extended verdict is now VERIFIED AFTER MINOR REVISION. No formula, assumption, inequality, conclusion, or proof is changed. All remaining P01 body text, including its historical DRAFT-status sentence, is preserved; the current P01 header and project manifest govern status.

P02 v1.1 is VERIFIED / AUDITED after A02's sole required correction E01: the boundary-continuity paragraph explicitly names the ordered pair $(s_-, s_+)$. A02's original verdicts are CORE MINOR REVISION and EXTENDED VERIFIED; closure is CORE VERIFIED AFTER MINOR REVISION, EXTENDED VERIFIED. The registered audit source is `audits/A02_synchronous_subspace_and_equilibria.tex`; only its supplied PDF reading copy is available and is included under the same basename. No theorem, proof, displayed formula, table, count, or assumption changed. Historical DRAFT wording in the P02 body is preserved; its header and the manifest govern current status. Compilation is not mathematical verification. The P03–P18 fragments remain empty statement slots. No numerical experiment is supplied. Earlier contract and planning fragments retain historical status wording. The literature phase is frozen. The immutable blueprint is archived separately and is not included in this document.

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# 1 Canonical model contract

The explicit initialization prompt is authoritative. This fragment specifies the main models; formulas in the archived blueprint remain research candidates.

## 1.1 Main demand-modulated model

For $i \in \mathbb{Z}_n$, define the vector field $F^{\mathrm{DM}}$ by

$$\dot{x}_i = ax_i - kx_i^2 + \frac{cx_{i-1}}{1+x_i} - \frac{gx_i}{1+x_{i+q}} - hx_ix_{i-q}. \tag{$\mathrm{DM}_{n,q}$}$$

The variables are dimensionless normalized activities. The preferred term is *demand-modulated model*. The positive input is called demand-modulated predecessor input; no standard Holling type-II classification is adopted. The intended initial conditions are $x(0) \in \mathbb{R}^n_{\geq 0}$. For local statements and boundary bifurcations, use the specified open extension domain $\mathcal{U} = (-1,\infty)^n$. This domain choice asserts no invariance theorem.

## 1.2 Primary matched bilinear comparator

Define $F^{\mathrm{MB}}$ by

$$\dot{y}_i = ay_i - ky_i^2 + cy_iy_{i-1} - gy_iy_{i+q} - hy_iy_{i-q}, \qquad i \in \mathbb{Z}_n. \tag{$\mathrm{MB}_{n,q}$}$$

This is the topology- and parameter-count-matched comparator in the main program. Its intended initial data are nonnegative; the polynomial vector field is specified on $\mathbb{R}^n$. No existence, invariance, boundedness or stability conclusion is included in this contract. Matching parameter names does not settle a quantitative calibration between the two laws.

## 1.3 Parameter and indexing contract

$$n \in \mathbb{Z}, \quad n \geq 5, \qquad q \in \{1,\dots,n-1\}, \qquad \boldsymbol{\pi} = (a,k,c,g,h) \in (0,\infty)^5.$$

All indices are modulo $n$, with representatives $0,\dots,n-1$. Arbitrary $q$ is allowed. The interpretation as one connected step-$q$ cycle requires $\gcd(n,q) = 1$; this condition is not imposed on the general analytical program. In particular, even $n$ is not excluded when $q = 2$. Coincident offsets do not remove terms from either defining equation. The intended exact wave-number-selection program focuses on $q = 2$. The five-node Wu-Xing-inspired specialization is $n = 5, q = 2$.

## 1.4 Archived secondary six-term comparator

The following is a transcription of the blueprint's proposed arbitrary-$n,q$ extension, not a verified attribution to the original Lin–Han publication:

$$\dot{y}_i = ay_i - by_i^2 + cy_iy_{i-1} - dy_iy_{i-q} - g_{\mathrm{c}}y_iy_{i+1} - hy_iy_{i+q}. \tag{$\mathrm{LB}_{n,q}$ archive}$$

Its archived five-node equation is obtained by specifying $n = 5, q = 2$ in this transcription. The six coefficient names are archival and are not a parameter identification with DM or MB. The original source, coefficient domain, index orientation, and attribution remain to be checked before analysis or use as a literature application.

### 1.5 Scope boundary

The main program covers homogeneous DM, primary comparator MB, their synchronous equilibria and spectra, the planned bifurcations, and the planned small-heterogeneity response theory. All requested P01–P18 tasks remain in the project; final main-text versus supplement placement is open. The heterogeneous coefficient assignment is not frozen by this contract. The six-term model is archived for a possible secondary literature application, never a replacement for MB. Source-saturating replacements, local motif theory, pathology extensions, adaptive set points, interventions, noise, delays, data calibration, and prior Lyapunov-spectrum reanalysis are postponed to other work. The master remains unchanged.

## 2 Notation and assumptions

### 2.1 Standing assumptions

**A1.** $n \in \mathbb{Z}$, $n \geq 5$, and $q \in \{1, \ldots, n-1\}$.

**A2.** Homogeneous parameters $\boldsymbol{\pi} = (a, k, c, g, h) \in (0, \infty)^5$.

**A3.** Dimensionless real activities; initial states in the nonnegative orthant when forward biological-admissibility statements are sought. For DM the local extension domain is $\mathcal{U} = (-1, \infty)^n$.

**A4, conditional.** Impose $\gcd(n, q) = 1$ only for the connected step-$q$ cycle interpretation; specify $q = 2$ or $(n, q) = (5, 2)$ when required by a module.

**A5, conditional.** Positivity of an equilibrium, invertibility, hyperbolicity, stability, Hopf nondegeneracy, and perturbation regularity are additional hypotheses to be stated and checked in the appropriate future proof. None is a standing established result.

### 2.2 Indices and cyclic shifts

Define $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$, using representatives $0, \ldots, n-1$. For $z \in \mathbb{C}^n$, define the predecessor and successor shift matrices by

$$(Pz)_j = z_{j-1}, \qquad (P^{-1}z)_j = z_{j+1}, \qquad P_{j\ell} = \begin{cases} 1, & \ell \equiv j-1 \pmod{n}, \\ 0, & \text{otherwise.} \end{cases}$$

Thus the symbols $P^q$ and $P^{-q}$ denote repeated predecessor and successor shifts, respectively. The group-action convention is $\rho(r)z = P^r z$. For transcription of the original five-node labels, the original $x_1, \ldots, x_5$ are relabeled as $x_0, \ldots, x_4$ in the same cyclic order. This is a label translation only; no historical or physiological statement is certified.

### 2.3 Fourier vectors, transform and pairing

Use i for the imaginary unit and $j$ for a node index. Define

$$\theta_m = \frac{2\pi m}{n}, \qquad v_j^{(m)} = \frac{1}{\sqrt{n}} \exp(\mathrm{i}\theta_m j), \qquad \mathsf{Q}_{jm} = v_j^{(m)}, \quad 0 \leq j, m < n.$$

The canonical transform convention is

$$\widehat{z}_m = \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} z_j \exp(-\mathrm{i}\theta_m j), \qquad z_j = \frac{1}{\sqrt{n}} \sum_{m=0}^{n-1} \widehat{z}_m \exp(\mathrm{i}\theta_m j).$$

The normalization is the unitary DFT convention. The blueprint's displayed unscaled Fourier vectors are not copied as normalized eigenvectors. Use $A^*$ for conjugate transpose, $A^{\mathsf{T}}$ for transpose, and

$$\langle u, v\rangle = u^* v = \sum_j \overline{u_j} v_j, \qquad \|z\|_2 = (z^* z)^{1/2}.$$

The pairing is conjugate-linear in its first argument. Matrix 2-norms are induced Euclidean norms. The adjoint eigenvector equation, its phase/scale, Taylor tensors and first-Lyapunov-coefficient convention must be reconciled in P13 before any blueprint normal-form formula is imported.

### 2.4 State sets and equilibrium notation

$$\mathbb{R}^n_{\geq 0} = \{x \in \mathbb{R}^n : x_j \geq 0 \text{ for all } j\}, \qquad \mathbb{R}^n_{>0} = \{x \in \mathbb{R}^n : x_j > 0 \text{ for all } j\},$$

$$\mathbf{1} = (1, \ldots, 1)^{\mathsf{T}}, \qquad \mathcal{S} = \{s\mathbf{1} : s \in \mathbb{R}\}.$$

The phrase positive orthant means the strict set $\mathbb{R}^n_{>0}$; nonnegative orthant means $\mathbb{R}^n_{\geq 0}$. A synchronous state is a member of $\mathcal{S}$; a positive synchronous equilibrium, if established, is denoted $x^*\mathbf{1}$ with $x^* > 0$. The admissible synchronous domain for DM is $\mathcal{S} \cap \mathcal{U}$. No invariance or equilibrium classification is asserted by these definitions.

### 2.5 Spectrum, hyperbolicity and mode terminology

For a square matrix $J$, define

$$\mathrm{sa}(J) = \max\{\mathrm{Re}\,\lambda : \lambda \in \mathrm{spec}(J)\}.$$

An equilibrium of a differentiable vector field is hyperbolic when its Jacobian has no eigenvalue with zero real part. A matrix is Hurwitz if all its eigenvalues have negative real part, and normal if $JJ^* = J^*J$. These definitions assert neither stability nor normality for a model. Mode $m = 0$ is called the uniform or synchronous mode; modes $m \neq 0$ are called nonuniform or symmetry-breaking modes. The mode-pair notation is $\{m, (-m) \bmod n\}$, usually written $\{m, n-m\}$. A pair representative is chosen in $0 \leq m \leq \lfloor n/2 \rfloor$. Keep the uniform mode and, for even $n$, $m = n/2$ as singleton labels. Any eigenvalue-conjugacy or simplicity claim belongs to P04 or P11. When a spectrum has been derived, the selected-mode set will mean $\arg\max_m \mathrm{Re}\,\lambda_m$, including every tie. A first unstable mode additionally requires a specified equilibrium branch and oriented parameter path.

### 2.6 Reserved scalar notation

For continuity with the archive, reserve $A = k + h$ and $\mu = a + c - g$ as abbreviations, without asserting an equilibrium equation or bifurcation. Use $\nu$ for a future scalar continuation parameter and $\varepsilon$ for a heterogeneity amplitude. Reserve $\omega > 0$ for a chosen positive Hopf frequency. Use $\mathrm{sa}(J)$ for spectral abscissa and $\delta = -\mathrm{sa}(J) > 0$ only under an explicitly stated Hurwitz hypothesis. This avoids reusing $\alpha$ as both a stability margin and a parameter perturbation. The spectral coefficient called $T_0$ in the blueprint will be named $\tau$ if needed, keeping $T_{\mathrm{scale}}$ available for a physical time scale. Heterogeneity profiles will be denoted collectively by $\boldsymbol{\zeta}$ if adopted; no coefficient-assignment convention is implied.

# 3 Theorem dependency map and completion gates

Arrows mean a prerequisite proof module, not a theorem already proved. Every module also depends on the canonical contract and notation. The following displayed adjacency graph is the full direct-dependency graph; transitive prerequisites are inherited. Conditional edges are listed separately.

$$
\begin{aligned}
\text{Contract} &\longrightarrow \text{P01}\\
\text{P01} &\longrightarrow \text{P02}\\
\text{P01, P02} &\longrightarrow \text{P03}\\
\text{P03} &\longrightarrow \text{P04}\\
\text{P04} &\longrightarrow \text{P05}\\
\text{P05} &\longrightarrow \text{P06}\\
\text{P05} &\longrightarrow \text{P07}\\
\text{P01, P02, P04, P05} &\longrightarrow \text{P08}\\
\text{P02, P03, P04} &\longrightarrow \text{P09}\\
\text{P02, P03, P04, P09} &\longrightarrow \text{P10}\\
\text{P02, P04, P05} &\longrightarrow \text{P11}\\
\text{P11} &\longrightarrow \text{P12}\\
\text{P11, P12} &\longrightarrow \text{P13}\\
\text{P11, P12, P13} &\longrightarrow \text{P14}\\
\text{P02, P04} &\longrightarrow \text{P15}\\
\text{P04, P15} &\longrightarrow \text{P16}\\
\text{P04, P16} &\longrightarrow \text{P17}\\
\text{P04, P12, P13, P15, P16} &\longrightarrow \text{P18}
\end{aligned}
$$

The graph is acyclic. P07 is an additional prerequisite when a five-node specialization is used in P11–P14. An optional periodic-orbit persistence extension in P15 would additionally require P14; it is not part of the core equilibrium-persistence gate. Source verification is a conditional input to any six-term extension in P08. An unresolved optional extension does not block the core module if its statement is explicitly excluded from the completed scope.

## 3.1 One dependency-respecting completion order

1. P01: Well-posedness and attracting bounded region.
2. P02: Synchrony and complete synchronous-equilibrium classification.
3. P03: Origin and homogeneous linearization.
4. P04: Arbitrary-(n,q) Fourier spectrum and normality.
5. P05: Exact step-two mode selection.
6. P06: Large-n selected-wave-number limit.
7. P07: Five-node golden-ratio specialization.
8. P08: Matched bilinear analysis and exact comparison.
9. P09: Transcritical bifurcation at the origin.
10. P10: Saddle-node bifurcation and true bistable region.

11. P11: Hopf spectral conditions and spectral simplicity.
12. P12: Hopf transversality.
13. P13: First Lyapunov coefficient and nondegenerate Hopf theorem.
14. P14: Equivariant rotating waves.
15. P15: Persistence under parameter heterogeneity.
16. P16: Static sensitivity and localized Green function.
17. P17: Transient and harmonic response.
18. P18: Stability radius and Hopf-threshold sensitivity.

After P05, P06, P07 and the later independent branches need not be worked in strict numerical order. This list is one valid order for the compendium and for eventual completion. P11 collects spectral conditions, P12 establishes transversality, and P13 is the home of the assembled nondegenerate Hopf theorem. P14 owns the additional equivariant rotating-wave conclusion.

### 3.2 Numerical dependency gates

| Task | Required analytical inputs before the claim is validated |
|---|---|
| N01 Jacobian and boundary checks | P01–P03; P08 for MB. |
| N01 equilibrium and spectrum checks | P02–P04; P05–P07 for selection, limits and specialization. |
| N01/N02 bifurcation validation | P09–P13; complete-spectrum, transversality and independent l1 checks. |
| N02 periodic branch and wave validation | P13–P14; use Floquet and phase checks, not a time-series image alone. |
| N01/N02 heterogeneity validation | P15–P16 for equilibrium/remainder; P18 for threshold shifts. |
| N01/N03 response and robustness | P16–P18; specify norms, forcing and direct versus equilibrium-induced perturbations. |
| N03 final figures | Each figure's listed proof owners, passed N01 checks and reproducible N02 output where relevant. |

Exploratory future checks may help find errors in DRAFT derivations, but cannot promote a theorem or a numerical claim to verified evidence on their own. No computational checks are performed in this initialization.

### 3.3 Manuscript claims withheld until verification

| Withheld claim | Minimum proof gate |
|---|---|
| Local well-posedness, positivity, global existence and bounded attraction | P01 with precise attraction/absorption language. |
| Complete synchronous-equilibrium classification and origin stability | P02–P03; nonlinear boundary cases handled separately. |
| Arbitrary-(n,q) spectrum and normality | P04. |
| Exact q=2 selection, large-n limit and five-node constants | P05, P06 and P07, respectively. |
| Exact matched-bilinear comparison | P08 plus its prerequisites and an explicit calibration for quantitative comparisons. |
| Transcritical, fold and true bistability claims | P09–P10 with all-mode stability and explicit basin scope. |
| Nondegenerate symmetry-breaking Hopf | P11–P13, including rigorous nonzero frequency, simplicity, transversality and l1 nonvanishing. |
| Stable rotating-wave branch | P14 and applicable transverse stability from P11–P13. |
| Persistence, sensitivity and localized response | P15–P16; positivity, invertibility and stability assumptions distinguished. |
| Transient/harmonic response and robustness | P17–P18, with explicit norms and remainder bounds. |
| Numerical values, periods, figures or observed regimes | Applicable verified proofs plus N01/N02 evidence and recorded reproduction. |
| Novelty, priority or literature comparisons | Completed source-specific literature audits; none confirmed here. |

Before submission, mathematical claims must also have a completed independent audit for the exact version cited. Unsupported candidate claims remain in the archive and registry. No final-paper narrative is supplied by this map.

### 3.4 Phase 1C frozen scope and current attempt order

The adjacency graph above remains in force. The numerical ordering above and in the compendium is a valid reading order. The current research attempt order, superseding earlier scheduling recommendations, is

$$\begin{gathered}\mathrm{P01} \to \mathrm{P02} \to \mathrm{P03} \to \mathrm{P04} \to \mathrm{P05} \to \mathrm{P08} \to \mathrm{P07} \to \mathrm{P09} \to \mathrm{P10},\\ \mathrm{P11} \to \mathrm{P12} \to \mathrm{P13} \to \mathrm{P14} \to \mathrm{P06},\\ \mathrm{P15} \to \mathrm{P16} \to \mathrm{P17} \to \mathrm{P18} \quad \text{(postponed extension queue)}.\end{gathered}$$

Arrows in this schedule indicate order of attempts, not additional mathematical dependencies. P06 is attempted only if it adds a useful large-$n$ consequence. P07 precedes any five-node use in P11–P14. P15–P18 are outside the frozen paper.

| Intended claim | Assembly gate | Frozen role |
|---|---|---|
| Realizable $q = 2$ selection and switching | P05 | Main theorem target; quadratic maximization alone is insufficient. |

| Intended claim | Assembly gate | Frozen role |
|---|---|---|
| Matched DM–MB contrast | P08; P13 if comparing criticality | Main theorem target on a common admissible branch domain. |
| First symmetry-breaking Hopf | P11, P12, P13 | Assembled main theorem; P11 alone is spectral support. |
| Nontrivial $l_1$ sign region | P13 | Main theorem target; currently unresolved. |
| Coefficient reduction | P13 | Proposition/lemma; detailed algebra may be supplementary. |
| Exact rotating-wave branch | P14 | Corollary of verified Hopf and symmetry hypotheses. |
| Five-node motivating application | P07 plus applicable P11–P14 | Corollary within the main paper. |
| General spectrum; branch classification and local branch bifurcations | P01–P04, P09–P10 | Supporting propositions/lemmas. |
| Useful large-$n$ consequence | P06 | Conditional corollary. |
| Heterogeneity, response and robustness | P15–P18 | Postpone. |

The primary comparison uses identical positive coefficient tuples with each model's own named positive synchronous branch; branch/path choices must be stated before their use. A criticality comparison is assembled after P13 and does not introduce a circular prerequisite for P08's spectral analysis.

The numerical gates above remain valid. Checks and figures for P15–P18 are dormant in this paper. No numerical task is executed in Phase 1C. All proof statuses remain TODO. Exact-model priority is not closed because the named 2021–2024 main-text equation/theorem inventories remain incomplete; this is a separate publication-claim gate, not a theorem dependency. See `frozen_theorem_package.md` and `final_novelty_verdict.md`.

## 4 Literature audit template

Copy this form for each source in a future audit. Keep an unfilled field unresolved; do not infer absence of overlap from absence of evidence.

**Record ID, reviewer, date and version:**

**Source:**

Full authors, title, year, venue, DOI or stable URL; primary source file and provenance; verification of bibliographic metadata.

**Exact model:**

Transcribed equations, parameter domain, normalization, node and edge conventions, boundary assumptions; source transcription check.

**Theorem or numerical result:**
Exact statement and evidence type; separate a proved theorem from a simulation or interpretation.

**Assumptions:**
All stated and implicit hypotheses, exclusions, nondegeneracy conditions, parameter restrictions and computational settings.

**Page/equation/theorem reference:**
Exact location in the consulted source, including version and supplement where relevant.

**Relation to our work:**
Corresponding registry IDs and proof modules; explicit dictionary between equations, parameters and indexing conventions.

**Possible overlap:**
Matching conclusions, proof methods or model classes; whether a specialization or direct corollary would cover our proposed result.

**Confirmed novelty:**
Leave empty until comparison is complete; attach evidence and reviewer sign-off for a narrowly stated novelty claim.

**Unresolved concern:**
Missing hypotheses, ambiguous attribution, unavailable source, inconsistent formula or unreproduced numerical claim.

**Disposition:**
Unresolved / source verified / overlap confirmed / qualified distinction documented. Record downstream claim and scope changes.

### 4.1 Future search clusters, not reviewed here

Cyclic feedback and circulant spectra; coupled-cell synchrony; equivariant Hopf and rotating waves; perturbation theory, heterogeneity and Green functions; exact Wu-Xing models and primary sources for interpretive language. No bibliography entry in the blueprint has been certified by this template.

# 5 P01: Well-posedness and bounded absorbing sets

## 5.1 Domain, hypotheses, and set terminology

Use exactly the demand-modulated vector field in $(\mathrm{DM}_{n,q})$, denoted here by $F = (F_i)_{i\in\mathbb{Z}_n}$. The canonical assumptions imply the weaker hypotheses

$$n \geq 1, \quad q \in \mathbb{Z} \pmod{n}, \quad a \in \mathbb{R}, \quad k > 0, \quad c, g, h \geq 0. \tag{P01-W}$$

We prove the main results under (P01-W), without changing the canonical contract for later modules. Coincident offsets retain every term of the defining equation. Write

$$\mathcal{U} = (-1, \infty)^n, \qquad K_r = [0, r]^n \ (r \geq 0), \qquad X(t) = \max_{i\in\mathbb{Z}_n} x_i(t), \qquad s_+ = \max\{s, 0\}.$$

For a continuous real function $v$, define

$$D^+ v(t) = \limsup_{\tau\downarrow 0} \frac{v(t+\tau) - v(t)}{\tau}.$$

Distances use the maximum norm; for $x \in \mathbb{R}^n_{\geq 0}$,

$$\mathrm{dist}_\infty(x, K_r) = (\max_i x_i - r)_+. \tag{1}$$

A set is *positively invariant* if solutions starting there remain there for all forward times for which they exist. It *absorbs bounded subsets of* $\mathbb{R}^n_{\geq 0}$ if every bounded $E \subset \mathbb{R}^n_{\geq 0}$ has a finite $T_E$ such that all solutions starting in $E$ lie in the set for every $t \geq T_E$. Attraction means convergence of the distance to zero; uniform attraction of bounded sets means convergence uniform over each such $E$. We will also establish stronger absorption bounds with a single time valid for the entire orthant.

## 5.2 Finite maxima, local solutions, and coordinate boundaries

**Lemma 5.1** (Finite maxima, including ties)**.** *For finitely many continuously differentiable functions $y_j$, set $v(t) = \max_j y_j(t)$ and $I(t) = \{j : y_j(t) = v(t)\}$. Then $v$ is locally Lipschitz, its right derivative exists at every time with a right neighborhood in the interval, and*

$$D^+ v(t) = \max_{j\in I(t)} \dot{y}_j(t). \tag{2}$$

*Proof.* Bounded derivatives on compact time intervals make the finite maximum Lipschitz. At a fixed $t$, every nonmaximizing index has a positive gap from $v(t)$. Finiteness and continuity ensure that none of these indices maximizes at $t + \tau$ for sufficiently small $\tau > 0$. Uniformly over $I(t)$, $y_j(t+\tau) = v(t) + \tau\dot{y}_j(t) + o(\tau)$. Taking the maximum and dividing by $\tau$ proves the assertion. □

**Proposition 5.2** (Local well-posedness and nonnegative invariance)**.** *For any finite real coefficients, $F$ is $C^\infty$ on $\mathcal{U}$. Every $x^0 \in \mathcal{U}$ has a unique maximal classical forward solution $x : [0, T_{\max}) \to \mathcal{U}$, with continuous dependence on initial data. If $c \geq 0$ and $x^0 \in \mathbb{R}^n_{\geq 0}$, then $x(t) \in \mathbb{R}^n_{\geq 0}$ for $t < T_{\max}$, regardless of the signs of the other coefficients.*

*Proof.* The denominators $1 + x_i$ and $1 + x_{i+q}$ are strictly positive on $\mathcal{U}$. Each component of $F$ is a sum of polynomials and rational functions with nonvanishing denominators, hence is smooth and locally Lipschitz. Standard local ODE existence, uniqueness, and maximal continuation theory on an open domain applies.

On each coordinate face,

$$x_i = 0, \quad x \in \mathbb{R}^n_{\geq 0} \quad \Longrightarrow \quad F_i(x) = cx_{i-1} \geq 0. \tag{3}$$

To treat tangent faces rigorously, fix $T < T_{\max}$ and let $x^+(t)$ be the coordinatewise positive part of $x(t)$. The curves $x, x^+$ and their connecting line segments lie in a compact rectangle in $\mathcal{U}$. Choose a Lipschitz constant $L$ for $F$ on this rectangle, in the maximum norm, and put

$$V(t) = \max\{0, -x_0(t), \ldots, -x_{n-1}(t)\} = \left\|x(t) - x^+(t)\right\|_\infty .$$

If $V > 0$ and $-x_i = V$, then $x_i^+ = 0$, so $F_i(x^+) \geq 0$ and $-F_i(x) \leq |F_i(x) - F_i(x^+)| \leq LV$. If $V = 0$, any active coordinate has $x_i = 0$ and $-F_i(x) \leq 0$; the constant zero is also active. Lemma 5.1 gives $D^+V \leq LV$. Since $V$ is Lipschitz, this inequality holds for its ordinary derivative almost everywhere. Integration of $(e^{-Lt}V)' \leq 0$, with $V(0) = 0$, gives $V = 0$ on $[0, T]$. The choice of $T$ was arbitrary. □

**Proposition 5.3** (Persistence and propagation of positivity)**.** *For $c \geq 0$ and nonnegative initial data, every initially positive coordinate remains strictly positive on $[0, T_{\max})$. In particular, strictly positive initial data stay strictly positive. If $c > 0$ and $x^0 \neq 0$, all coordinates are strictly positive at every $0 < t < T_{\max}$. If $c = 0$, initially zero coordinates remain zero. The zero initial state remains identically zero.*

*Proof.* Along a nonnegative solution write

$$\dot{x}_i = b_i(t)x_i + d_i(t), \qquad b_i = a - kx_i - \frac{g}{1 + x_{i+q}} - hx_{i-q}, \qquad d_i = \frac{cx_{i-1}}{1 + x_i} \geq 0.$$

These coefficients are continuous on every compact time interval in the maximal interval. The integrating-factor formula gives

$$x_i(t) = x_i^0 e^{\int_0^t b_i(s)\,ds} + \int_0^t e^{\int_s^t b_i(r)\,dr} d_i(s)\,ds. \tag{4}$$

This proves persistence of an initially positive coordinate. If $c > 0$, a predecessor positive at every $s > 0$ supplies a strictly positive integrand to its successor at every $s > 0$, making that successor positive at every $t > 0$. At most $n - 1$ iterations along the step-one cycle reach all nodes, independently of $q$. For $c = 0$ the integral vanishes, proving preservation of zero coordinates. Finally $F(0) = 0$ and uniqueness proves the zero-data assertion. □

### 5.3 Maximum comparison and the exact comparison radius

Under (P01-W), define

$$b(s) = a - ks + \frac{c}{1 + s}, \qquad H(s) = sb(s) \quad (s \geq 0),$$

and set

$$X_{\max} = R := \max\left\{0, \frac{a - k + \sqrt{(a + k)^2 + 4kc}}{2k}\right\}. \tag{5}$$

This radius is defined by a scalar upper comparison; it is not asserted to be the smallest invariant cube of the full vector field.

**Lemma 5.4** (Dini inequality and scalar root)**.** *Every nonnegative solution under* (P01-W) *satisfies*

$$D^+X = \max_{i:x_i=X} F_i(x) \le H(X) - \frac{gX}{1+X} \le H(X). \tag{6}$$

*If $a+c>0$, then $R>0$ is the unique positive zero of $b$, and*

$$kR^2 + (k-a)R - (a+c) = 0. \tag{7}$$

*If $a+c\le 0$, then $R=0$ and $H(s)\le -ks^2$ for $s\ge 0$. For $R>0$ and $s\ge R$,*

$$H(s) = s\left[-k(s-R) + c\left(\frac{1}{1+s} - \frac{1}{1+R}\right)\right] \le -ks(s-R). \tag{8}$$

*Proof.* Lemma 5.1 includes every tied maximizing coordinate. At $x_i = X$, nonnegative invariance gives $0 \le x_{i-1}, x_{i+q}, x_{i-q} \le X$. Consequently

$$\frac{cx_{i-1}}{1+X} \le \frac{cX}{1+X}, \qquad -\frac{gX}{1+x_{i+q}} \le -\frac{gX}{1+X}, \qquad -hXx_{i-q} \le 0.$$

The middle inequality uses the negative prefactor and $g \ge 0$. Adding proves (6); no independence of neighbor coordinates is required.

Now $b'(s) = -k - c/(1+s)^2 < 0$, $b(0) = a+c$, and $b(s) \to -\infty$. If $a+c>0$, multiply $b(R)=0$ by $1+R$ to obtain (7). Its constant term is negative and its discriminant is $(a+k)^2 + 4kc$, so its positive root is (5). If $a+c\le 0$, then $k-a>0$ and

$$(a+k)^2 + 4kc - (k-a)^2 = 4k(a+c) \le 0.$$

Thus the raw root in (5) is nonpositive, $R=0$, and $b(s) \le a+c-ks \le -ks$. Subtracting $b(R)=0$ when $R>0$ gives (8); its inequality follows from $c\ge 0$ and $s\ge R$. □

**Theorem 5.5** (Global continuation and invariant attracting cubes)**.** *Under* (P01-W)*, every $x^0 \in \mathbb{R}^n_{\ge 0}$ has a unique classical solution for all $t\ge 0$. The scalar solution*

$$\dot{u} = H(u), \qquad u(0) = X(0) \tag{9}$$

*is global and nonnegative, and*

$$0 \le X(t) \le u(t) \le \max\{X(0), R\}. \tag{10}$$

*Every $K_r$ with $r\ge R$ is positively invariant. Furthermore,*

$$\operatorname{dist}_\infty(x(t), K_R) \le e^{-kRt}(X(0)-R)_+ \qquad (R>0), \tag{11}$$

$$X(t) \le \frac{X(0)}{1+kX(0)t} \qquad (R=0). \tag{12}$$

*There are also initial-state-independent bounds, for $t>0$:*

$$X(t) \le \frac{R}{1-e^{-kRt}} \qquad (R>0), \tag{13}$$

$$X(t) \le \frac{1}{kt} \qquad (R=0). \tag{14}$$

*Thus $K_R$ attracts all nonnegative solutions uniformly on bounded initial sets, and indeed uniformly over the entire orthant as $t\to\infty$. Every larger cube $K_r$, $r>R$, absorbs the entire orthant by the common time*

$$T_r = \begin{cases} \dfrac{1}{kR}\log\dfrac{r}{r-R}, & R>0, \\ \dfrac{1}{kr}, & R=0. \end{cases} \tag{15}$$

*Proof.* The scalar field $H$ is smooth on $(-1,\infty)$, $H(0)=0$, and $H(s)\le 0$ for $s\ge R$. Its solution starting nonnegative stays in $[0,M]$, where $M=\max\{u(0),R\}$: uniqueness prevents crossing zero, and the upper endpoint has nonpositive field. The latter barrier also follows by the positive-part argument below applied to $(u-M)_+$, using $H(M)\le 0$. Compact containment gives global scalar continuation.

We justify the vector comparison without assuming differentiability of $X$ everywhere. On a compact subinterval of the vector maximal interval, let $L$ be a Lipschitz constant for $H$ on an interval containing $X$ and $u$. The Lipschitz function

$$W=(X-u)_+=\max\{0,x_i-u : i\in\mathbb{Z}_n\}$$

satisfies $D^+W\le H(X)-H(u)\le LW$ where $W>0$. At $W=0$, any active coordinate has $x_i=u=X$, unless the constant zero is the only active function; Lemma 5.1 and (6) give $D^+W\le 0$. Integrating the almost-everywhere inequality, as in Proposition 5.2, yields $W=0$. This proves (10) on the maximal interval.

If $T_{\max}<\infty$, the vector trajectory stays in $K_M$, a compact subset of $\mathcal{U}$ at distance at least one from every singular hyperplane. The field is bounded on a compact neighborhood in $\mathcal{U}$, hence $\dot{x}$ is bounded. Thus $x(t)$ has a limit in $K_M$ as $t\uparrow T_{\max}$. Local existence at that limit and uniqueness extend the solution past $T_{\max}$, a contradiction. This proves global continuation and absence of finite-time blow-up. If $X(0)\le r$ and $r\ge R$, (10) proves invariance.

For $R>0$, (8) gives $\dot{u}\le -ku(u-R)\le -kR(u-R)$ where $u\ge R$. It follows that $(u(t)-R)_+\le e^{-kRt}(u(0)-R)_+$, including solutions initially at or below $R$, and hence (11). If $u(0)>R$, set $w=1-R/u$ while $u>R$. Then

$$\dot{w}=\frac{R\dot{u}}{u^2}\le -kRw,\qquad 0<w(0)<1.$$

Solving this inequality gives (13) for $u$ and therefore for $X$. If $u$ reaches $R$, invariance supplies the same bound afterwards; if $u(0)\le R$, it is immediate. For $R=0$, integrating $(1/u)'\ge k$ where $u>0$, using $\dot{u}\le -ku^2$, proves (12) and (14); zero initial data follow by uniqueness.

Attraction follows from these estimates and (1). For $R>0$ the distance is at most $R/(e^{kRt}-1)$ independently of initial data; for $R=0$ use $1/(kt)$. Substitution of $T_r$ into (13) or (14) gives $X(T_r)\le r$, and invariance supplies all subsequent times. □

### 5.4 Finite-time absorption at the comparison radius

**Theorem 5.6** (Exact absorption criterion for $K_R$)**.** *Assume* (P01-W). *If $R=0$, no initially nonzero nonnegative solution enters $K_R=\{0\}$ at finite time. This cube is invariant and globally attracting but not absorbing.*

*If $R>0$, the following are equivalent:*

*(i) $K_R$ absorbs every nonnegative solution in finite time.*

*(ii) $K_R$ absorbs bounded subsets of $\mathbb{R}^n_{\ge 0}$.*

*(iii) One finite absorption time is valid for the entire orthant.*

*(iv) The coefficients and offset satisfy*

$$g>0 \quad or \quad [\,h>0 \text{ and } (c>0 \text{ or } q\equiv 0 \pmod n)\,]. \tag{16}$$

*Canonically $R > 0$, $g > 0$, and the entire orthant enters $K_R$ by*

$$T_* = \frac{\log 2}{kR} + \frac{1+R}{g}. \tag{17}$$

*If $g > 0$ and $X(0) \le M$, an alternative entry-time bound is*

$$T_M = \frac{(M-R)_+(1+R)}{gR}. \tag{18}$$

*Proof.* When $R = 0$, Proposition 5.3 ensures that some coordinate of any initially nonzero solution stays positive at each finite time.

Now suppose $R > 0$ and set $t_0 = (\log 2)/(kR)$. By (13), every trajectory is in the invariant cube $K_{2R}$ by $t_0$. If $g > 0$, then whenever $X \ge R$,

$$D^+X \le -\frac{gX}{1+X} \le -\frac{gR}{1+R} =: -\delta < 0, \tag{19}$$

using $H(X) \le 0$. Absolute continuity permits integration before the first entry into $X \le R$. If $X(0) \le M$, a longer time outside than $(M-R)_+/\delta$ would force $X \le R$, a contradiction. Invariance gives all later times. Applying this after $t_0$ with $M = 2R$ proves (17).

If $g = 0$, $h > 0$, and $q \equiv 0 \pmod n$, a maximizing coordinate has $x_{i-q} = X$. Thus $D^+X \le H(X) - hX^2 \le -hR^2$ when $X \ge R$, giving common absorption by $t_0 + 1/(hR)$.

For the remaining sufficient case $g = 0$, $h > 0$, $c > 0$, we give a coordinatewise quantitative argument. Put $M = 2R$ and fix any $\tau > 0$. On $K_M$,

$$\dot{x}_i \ge -\beta x_i + \eta x_{i-1}, \qquad \beta = (-a)_+ + (k+h)M + g, \quad \eta = \frac{c}{1+M} > 0. \tag{20}$$

Indeed $a \ge -(-a)_+$, $kx_i \le kM$, $g/(1+x_{i+q}) \le g$, $hx_{i-q} \le hM$, and $c/(1+x_i) \ge c/(1+M)$. On $[s, s+\tau]$ in this cube, the transformed variables $z_i(r) = e^{\beta(r-s)}x_i(r)$ satisfy

$$z_i(r) \ge x_i(s) + \eta \int_s^r z_{i-1}(v)\, dv.$$

Repeated substitution, keeping the remaining nested integral nonnegative, yields

$$x_i(s+\tau) \ge e^{-\beta\tau} \sum_{j=0}^{n-1} \frac{(\eta\tau)^j}{j!} x_{i-j}(s). \tag{21}$$

A $j$-fold integral of a constant over the time simplex equals $\tau^j/j!$, which gives the displayed coefficients.

For a trajectory that has not entered $K_R$ by time $t \ge t_0 + \tau$, we have $X(t-\tau) > R$. The indices $i - j$, $0 \le j \le n-1$, cover the cycle once. Thus (21) gives, for every $i$,

$$x_i(t) \ge d := Re^{-\beta\tau} \min_{0 \le j \le n-1} \frac{(\eta\tau)^j}{j!} > 0. \tag{22}$$

At such times, every maximizing coordinate satisfies $F_i(x) \le H(X) - hXx_{i-q} \le -hRd$. Starting from $X \le 2R$ at $t_0 + \tau$, integration proves entry by $t_0 + \tau + 1/(hd)$, or earlier. We have proved (iv)⇒(iii).

The implications (iii)⇒(ii)⇒(i) are immediate. If (iv) fails and $g = h = 0$, choose all initial coordinates equal to $s_0 > R$. Direct substitution verifies the solution $x_i(t) = s(t)$ with $\dot{s} = H(s)$. It

stays above $R$ at each finite time: a first meeting with the constant scalar solution $R$ would violate uniqueness, also backwards near that meeting. Hence it never enters $K_R$. The other failure case is $g = 0$, $h > 0$, $c = 0$, $q \not\equiv 0 \pmod{n}$. Here $R = a/k > 0$. Choose one initial coordinate $x_j(0) > R$ and all others zero. The zero coordinates stay zero and $j - q \neq j$, so $\dot{x}_j = ax_j - kx_j^2$. It likewise stays strictly above $a/k$ at every finite time. These cases exhaust failure of (iv) and contradict (i), completing the equivalence. □

### 5.5 Canonical conclusions and weaker-assumption limits

**Corollary 5.7** (Canonical foundation)**.** *For canonical $n \geq 5$, $1 \leq q \leq n-1$ and $a, k, c, g, h > 0$, every nonnegative initial state has a unique global nonnegative solution. Every initially positive coordinate stays positive, and every nonzero initial state becomes strictly positive in every coordinate at each positive time. The cube $[0, X_{\max}]^n$, with*

$$X_{\max} = \frac{a - k + \sqrt{(a+k)^2 + 4kc}}{2k} > 0,$$

*is positively invariant, globally attracting, and finite-time absorbing for the entire orthant with the common bound* (17)*. All larger cubes are also invariant and absorbing. No condition on* $\gcd(n, q)$ *is required.*

*Proof.* The canonical assumptions imply (P01-W), $a + c > 0$, $c > 0$ and $g > 0$. Apply the preceding results. □

*Remark* 5.8 (Archived radius and the meaning of stability). The blueprint's candidate radius

$$B = \frac{a + \sqrt{a^2 + 4kc}}{2k}$$

is the larger real root of $as - ks^2 + c = 0$, and is nonnegative. If $c > 0$, then $B > 0$ and $H(B) = -c/(1 + B) < 0$, so $B > R$. Its cube is therefore also invariant and absorbing by Theorem 5.5, even when $g = h = 0$. The smaller radius $R$ retains $cX/(1+X)$ instead of replacing it by $c$. If $c = 0$, then $B = R$ and the exact-radius criterion applies. Neither radius is claimed optimal after all $g, h, q$ information is used.

Invariance concerns remaining inside; absorption concerns finite-time entry; attraction alone permits approaching from outside forever. The distance estimates also show Lyapunov stability of $K_R$ as a set, relative to $\mathbb{R}^n_{\geq 0}$, because distance never exceeds its initial value. Under the convention that asymptotic stability of a set means Lyapunov stability plus attraction, $K_R$ is globally asymptotically stable as a set. This is not convergence to a particular point, nor a claim that the cube is a minimal global attractor or is invariant in both time directions.

**Proposition 5.9** (Global existence when $k = 0$ is allowed)**.** *Global nonnegative existence and the positivity conclusions still hold for $k \geq 0$, $a \in \mathbb{R}$, $c, g, h \geq 0$, any $n \geq 1$, and any residue $q$. For $k = 0$, no bounded absorbing cube is guaranteed over this parameter class.*

*Proof.* Local existence and nonnegative invariance did not require $k > 0$. Discarding nonpositive losses at a maximizing coordinate gives

$$D^+ X \leq aX + \frac{cX}{1+X} \leq (a_+ + c)X.$$

The Lipschitz/Dini argument yields $X(t) \leq X(0)e^{(a_+ + c)t}$. If the maximal time were finite, this would confine the trajectory to a compact subset of $\mathcal{U}$, and the earlier continuation argument would extend it. Positivity follows from Proposition 5.3. For the last assertion take $a > 0$, $k = c = g = h = 0$: an initially positive coordinate equals $x_i(0)e^{at}$ and is unbounded. □

### 5.6 Checks and possible failure points

1. **Boundary tangency.** The negative-part maximum proves invariance even when the derivative on a face vanishes. A bare first-contact derivative assertion is insufficient. If $c < 0$ and $n \geq 2$, a face state with $x_i = 0$, $x_{i-1} > 0$ has $\dot{x}_i < 0$, so invariance can fail. No forward assertion concerns initial data outside $\mathbb{R}^n_{\geq 0}$.

2. **Tied maxima and loss signs.** All active indices occur in (2). The negative prefactor in the $g$ term reverses the reciprocal inequality. Coincident offsets are harmless because the estimates do not optimize neighbors independently.

3. **Root degeneracy.** Equation (7) characterizes the positive comparison root. If $a + c \leq 0$, the selected radius is zero and need not be a root of that quadratic: the factor $s$ in $H(s)$ supplies the zero. No expression containing $1/R$ is used when $R = 0$.

4. **Absorption mechanism.** The equality $H(R) = 0$ alone does not imply finite-time entry. Canonical absorption uses the retained $g$ loss; when it vanishes, the exact criterion supplies another strict mechanism or a nonabsorbing trajectory. Positivity prevents finite-time entry into $K_0$ from nonzero data.

5. **Dimension and connectivity.** Nothing requires $n \geq 5$, coprimality, or distinct offsets. Positivity propagates through the step-one input when $c > 0$. The $q \equiv 0$ absorption case belongs only to the weaker extension of the canonical range. Strict positivity at each finite time is not a uniform positive lower bound as $t \to \infty$.

6. **Coefficient edges.** Strict $a, c, g, h > 0$ is unnecessary for the bounded-region theorem when $k > 0$; even $a \in \mathbb{R}$ is allowed. The condition $k \geq 0$ suffices for global existence. If $k < 0$ and $a = c = g = h = 0$, then $\dot{x}_i = |k| x_i^2$ blows up in finite time for a positive coordinate. If $h < 0$, the dissipative bounds also fail in general: take $k > 0$, $a = c = g = 0$, $h < -k$ and equal positive initial coordinates. Direct substitution gives $\dot{s} = -(k + h)s^2$ and finite-time blow-up. These examples are outside the proved domain, not necessity claims for every signed parameter choice.

7. **Continuation and status.** Boundedness is used together with compact containment in the smooth open domain, excluding denominator singularities. All maximum, comparison, and coordinate integral inequalities are proved here. Only standard local ODE existence/uniqueness/continuation theory is invoked. This module remains DRAFT pending a separate independent audit; no numerical experiment or later proof-module result is used.

# 6 P02: Synchrony and synchronous equilibria

**Status: DRAFT; independent audit pending.** The standing assumptions are the canonical A1–A3. In particular, $n \geq 5$, $1 \leq q \leq n-1$, $a, k, c, g, h > 0$, and the open extension domain is $\mathcal{U} = (-1, \infty)^n$. The nonnegative model state space is $\mathbb{R}^n_{\geq 0}$. P01 v1.1 is VERIFIED / AUDITED: its required wording correction E01 is already incorporated. Only its local uniqueness and, for nonnegative initial data, global continuation are needed here. The calculations below start from $(\mathrm{DM}_{n,q})$. No assertion about nonsynchronous equilibria, equilibrium stability, or bifurcation type is made.

## 6.1 Invariant synchronous subspace

**Proposition 6.1** (Invariance of the synchronous subspace)**.** *The set*

$$\mathcal{S} \cap \mathcal{U} = \{s\mathbf{1} : s \in (-1, \infty)\}$$

*is invariant for the canonical vector field for the entire maximal solution interval in $\mathcal{U}$. Its scalar restriction is*

$$\dot{s} = f(s) := as - ks^2 + \frac{cs}{1+s} - \frac{gs}{1+s} - hs^2, \qquad s \in (-1, \infty). \tag{23}$$

*This conclusion needs no coprimality or distinct-offset assumption. Indeed, the substitution and local invariance remain valid for any integer $n \geq 1$, any residue $q$ modulo $n$, and any finite real coefficients. The canonical restrictions are retained for the equilibrium classification below.*

*Proof.* Put $x = s\mathbf{1}$ with $s > -1$. For every $i \in \mathbb{Z}_n$, each of $x_i, x_{i-1}, x_{i+q}, x_{i-q}$ equals $s$. Direct substitution gives

$$F_i(s\mathbf{1}) = as - ks^2 + \frac{cs}{1+s} - \frac{gs}{1+s} - hs^2 = f(s).$$

Every component is identical, so $F(s\mathbf{1}) = f(s)\mathbf{1}$. Coincident offsets retain all the displayed terms; no distinctness of neighbors is used. Nor is the size or connectedness of a step-$q$ cycle used.

The scalar field is smooth on $(-1, \infty)$. A scalar solution $s(t)$ lifts to the vector solution $s(t)\mathbf{1}$. Uniqueness from Proposition 5.2 makes this lift agree with the vector solution having initial value $s(0)\mathbf{1}$ wherever both exist. If the scalar maximal interval ended earlier while the vector solution continued in $\mathcal{U}$, the latter would supply a limit $s_*\mathbf{1} \in \mathcal{U}$ at that time. Local scalar continuation at $s_* > -1$ would extend the lift, a contradiction. Conversely, a longer scalar interval would extend the vector solution. Thus the maximal intervals coincide and synchrony persists throughout. The same local uniqueness argument applies on either side of an initial time. The scalar domain inherited from $\mathcal{U}$ is exactly $(-1, \infty)$. □

## 6.2 Scalar synchronous dynamics

Use the reserved abbreviations and one local auxiliary abbreviation

$$A = k + h > 0, \qquad \mu = a + c - g, \qquad b = a - A. \tag{24}$$

Here $b$ is used only within P02; it is not the archived comparator's coefficient. In particular $b > -A$ because $a > 0$. The exact restriction is

$$f(s) = s\left(a - As + \frac{c-g}{1+s}\right) = \frac{s}{1+s}\big(\mu + bs - As^2\big), \qquad s > -1. \tag{25}$$

For $s(0) \geq 0$, P01 and Proposition 6.1 give a global nonnegative scalar solution. For $-1 < s(0) < 0$, only the maximal interval within this open domain is asserted. Even when $c = g$ cancels the two rational terms along synchrony, this does not enlarge the canonical domain or admit $s = -1$.

### 6.3 The origin and the nonzero equilibrium equation

**Proposition 6.2** (Equilibrium equation with domain restrictions)**.** *The origin* $0\mathbf{1}$ *is an equilibrium for every canonical parameter tuple. Every other synchronous equilibrium in* $\mathcal{U}$ *is exactly* $s\mathbf{1}$*, where* $s \neq 0$*,* $s > -1$*, and*

$$\begin{aligned} 0 = Q(s) &:= As^2 - bs - \mu \\ &= (k+h)s^2 + (k+h-a)s - (a+c-g). \end{aligned} \tag{26}$$

*Define the discriminant*

$$\begin{aligned} D &:= b^2 + 4A\mu \\ &= (a-A)^2 + 4A(a+c-g) = (a+A)^2 + 4A(c-g). \end{aligned} \tag{27}$$

*The complete list of real polynomial roots, when* $D \geq 0$*, is*

$$s_\pm = \frac{b \pm \sqrt{D}}{2A} = \frac{a-(k+h) \pm \sqrt{(a+k+h)^2 + 4(k+h)(c-g)}}{2(k+h)}. \tag{28}$$

*They are distinct for* $D > 0$ *and coincide for* $D = 0$*. When* $D < 0$ *there are no real roots of* $Q$*.*

*Proof.* Directly in (23), all five terms vanish at $s = 0$. Thus the origin is always an equilibrium, before any division is performed. Now take $s \neq 0$ with $s > -1$. Dividing (25) by $s$ and multiplying by the strictly positive number $1 + s$ gives

$$(a - As)(1+s) + c - g = \mu + (a-A)s - As^2 = 0.$$

Both operations are reversible on this stated set, giving (26). Completing the square yields

$$Q(s) = A\left(s - \frac{b}{2A}\right)^2 - \frac{D}{4A}. \tag{29}$$

Expansion of $b^2 + 4A\mu$ proves both discriminant forms and the quadratic formula gives (28). Finally,

$$Q(-1) = A + b - \mu = g - c. \tag{30}$$

Thus $s = -1$ is a polynomial root precisely when $g = c$, but it is never a point of the canonical domain. Multiplication by $1 + s$ does not authorize retaining that root. A root $s = 0$ of $Q$, when present, is the already counted origin rather than another equilibrium. □

### 6.4 Complete classification in the canonical parameter regime

In this subsection, a *mathematical root* means a real root of $Q$, including a zero or an inadmissible root. A *DM-domain root* additionally satisfies $s > -1$. A *nonnegative model equilibrium* has $s \geq 0$, and a *positive model equilibrium* has $s > 0$. The ever-present origin is added once to the admissible nonzero roots. All counts below count distinct values, not algebraic multiplicity.

**Theorem 6.3** (Complete synchronous-equilibrium classification)**.** *Under the canonical assumptions, the synchronous equilibrium set is*

$$\{0\mathbf{1}\} \ \cup \ \{s\mathbf{1} : Q(s) = 0,\ s > -1,\ s \neq 0\}. \tag{31}$$

*The following eleven cases are exhaustive. In the table,* $r$ *counts real nonzero polynomial roots,* $d$ *counts those with* $s > -1$*,* $N_0$ *counts all nonnegative synchronous equilibria (including the origin), and* $N_+$ *counts the strictly positive ones. The total number of synchronous equilibria in* $\mathcal{U}$ *is* $1 + d$*.*

| *Case* | *Parameter conditions* | *Real roots of $Q$ and location* | $r$ | $d$ | $N_0$ | $N_+$ |
|---|---|---|---|---|---|---|
| *I* | $\mu > 0,\ g < c$ | $s_- < -1 < 0 < s_+$ | *2* | *1* | *2* | *1* |
| *II* | $\mu > 0,\ g = c$ | $s_- = -1,\ s_+ = a/A > 0$ | *2* | *1* | *2* | *1* |
| *III* | $\mu > 0,\ g > c$ | $-1 < s_- < 0 < s_+$ | *2* | *2* | *2* | *1* |
| *IV* | $\mu = 0,\ b > 0$ | $s_- = 0,\ s_+ = b/A > 0$ | *1* | *1* | *2* | *1* |
| *V* | $\mu = 0,\ b = 0$ | $s_- = s_+ = 0$ *(double)* | *0* | *0* | *1* | *0* |
| *VI* | $\mu = 0,\ b < 0$ | $-1 < s_- = b/A < 0 = s_+$ | *1* | *1* | *1* | *0* |
| *VII* | $\mu < 0,\ D < 0$ | *No real roots; any sign of $b$* | *0* | *0* | *1* | *0* |
| *VIII* | $\mu < 0,\ D > 0,\ b > 0$ | $0 < s_- < s_+$ | *2* | *2* | *3* | *2* |
| *IX* | $\mu < 0,\ D = 0,\ b > 0$ | $s_- = s_+ = b/(2A) > 0$ | *1* | *1* | *2* | *1* |
| *X* | $\mu < 0,\ D > 0,\ b < 0$ | $-1 < s_- < s_+ < 0$ | *2* | *2* | *1* | *0* |
| *XI* | $\mu < 0,\ D = 0,\ b < 0$ | $-\frac{1}{2} < s_- = s_+ = b/(2A) < 0$ | *1* | *1* | *1* | *0* |

*When $\mu < 0$ and $b = 0$, necessarily $D = 4A\mu < 0$, so case VII applies. Each appearance of $s_\pm$ refers to* (28)*. The only repeated polynomial roots are in cases V, IX and XI.*

*As zeros of the original smooth scalar field $f$ on $(-1, \infty)$, all nonzero equilibria are simple except the double roots in IX and XI. The origin is simple if $\mu \neq 0$, double if $\mu = 0$, $b \neq 0$, and triple if $\mu = b = 0$. The algebraic continuation of the nonzero equilibrium curve meets the origin exactly at $\mu = 0$: through $s_-$ in case IV, through $s_+$ in case VI, and through both coincident root labels in case V. No negative root represents a state in $\mathbb{R}^n_{\geq 0}$.*

*Proof.* Proposition 6.2 proves completeness of (31). The Vieta relations are

$$s_- + s_+ = \frac{b}{A}, \qquad s_- s_+ = -\frac{\mu}{A}. \tag{32}$$

We first establish the domain test for every real pair of roots. Because $a > 0$ and $A > 0$,

$$s_+ \geq \frac{b}{2A} = \frac{a}{2A} - \frac{1}{2} > -\frac{1}{2} > -1.$$

For the smaller root,

$$s_- > -1 \iff \sqrt{D} < a + A \iff D < (a + A)^2 \iff g > c. \tag{33}$$

The middle equivalence uses $a + A > 0$ and $D \geq 0$. Equality gives $s_- = -1$ exactly when $g = c$; the reverse inequality gives $s_- < -1$ exactly when $g < c$. These are polynomial-root statements on the real-discriminant set, not assertions that such roots are admissible.

If $\mu > 0$, then $D > b^2$, so $s_- < 0 < s_+$ and both roots are nonzero and simple. The domain test gives I–III. When $g = c$, $D = (a + A)^2$ and the stated roots $-1, a/A$ follow directly.

If $\mu = 0$, then $Q(s) = s(As - b)$. If $b \neq 0$, its distinct roots are 0 and $b/A$. Since $b/A = a/A - 1 > -1$, the latter is always domain-admissible and has the sign of $b$. Their ordering gives IV and VI. If $b = 0$, $Q(s) = As^2$ gives V. The origin is counted only once in the equilibrium set in all three cases.

If $\mu < 0$, real roots exist precisely when $D \geq 0$. For $D < 0$ we obtain VII. For $D \geq 0$, the product in (32) is strictly positive and $b \neq 0$ (otherwise $D < 0$). Thus both roots are nonzero with the sign of their sum, namely the sign of $b$. They are distinct for $D > 0$ and double for $D = 0$. Also $g > a + c > c$, so both are admissible by (33). This gives VIII–XI, including the sharper negative repeated-root bound from $b > -A$. These arguments yield every count in the table.

On $s > -1$, the identity $f(s) = -sQ(s)/(1 + s)$ shows that the nonzero zeros have the same multiplicity as their polynomial roots. At the origin, use (25) without dividing: the numerator is

$\mu s + bs^2 - As^3$ and $1 + s$ does not vanish. This gives the stated orders one, two and three. Finally $Q(0) = -\mu$, so the nonzero equation's algebraic continuation contains $s = 0$ exactly when $\mu = 0$. The formulas at that locus give its root labels, and the local branch description below confirms the approach by nonzero roots. The counting and multiplicity assertions are purely algebraic and carry no stability conclusion. □

### 6.5 Positive synchronous equilibria: necessary and sufficient conditions

**Theorem 6.4** (Exact positive-equilibrium alternatives)**.** *In the canonical parameter regime the following four alternatives partition parameter space:*

*(i) There is no positive synchronous equilibrium if and only if*

$$[\,\mu = 0,\ b \le 0\,] \quad \text{or} \quad [\,\mu < 0,\ (b \le 0 \text{ or } D < 0)\,]. \tag{34}$$

*(ii) There is exactly one positive synchronous equilibrium and it is simple if and only if $\mu > 0$, or $\mu = 0$ and $b > 0$. It is $s_+\mathbf{1}$; in the second case $s_+ = b/A$.*

*(iii) There are exactly two distinct positive synchronous equilibria if and only if $\mu < 0$, $b > 0$, and $D > 0$. They are $s_-\mathbf{1}, s_+\mathbf{1}$, both simple.*

*(iv) There is exactly one positive synchronous equilibrium and it is repeated if and only if $\mu < 0$, $b > 0$, and $D = 0$. It is a double root, $s_-\mathbf{1} = s_+\mathbf{1} = (b/(2A))\mathbf{1}$.*

*Consequently, exactly one distinct positive equilibrium (without a simplicity qualification) occurs if and only if (ii) or (iv) holds. There are never more than two positive synchronous equilibria.*

*Proof.* The cases of Theorem 6.3 prove every implication, including the necessity and all equalities. One can also reconstruct the positive-root count directly from $H(s) = \mu + bs - As^2$ for $s > 0$. If $b \le 0$, then $H'(s) = b - 2As < 0$ on this interval: it has one positive zero exactly when $\mu > 0$. If $b > 0$, its maximum on $s \ge 0$ is at $s = b/(2A) > 0$ and has value $D/(4A)$. For $\mu < 0$, that maximum is positive, zero, or negative precisely in alternatives (iii), (iv), or (i). For $\mu = 0$, the zero at the endpoint is excluded from the positive count and the other root is $b/A > 0$. For $\mu > 0$, the opposite-sign Vieta product leaves one positive root. This gives an independent sign-and-vertex justification of the list. □

### 6.6 Algebraic branch loci and degeneracies

**Proposition 6.5** (Branch-intersection and repeated-root loci)**.** *For fixed positive $a, k, c, h$, define*

$$g_f := a + c + \frac{(a-A)^2}{4A} = c + \frac{(a+A)^2}{4A}, \qquad s_f := \frac{a-A}{2A}. \tag{35}$$

*The following statements concern only the scalar equilibrium equation.*

*(i) The* branch-intersection locus *is $\mu = 0$, or $g = a + c$. It is the set where the algebraic continuation of the nonzero equilibrium equation meets the origin branch.*

*(ii) The* repeated-root locus *of $Q$ is $D = 0$, or $g = g_f$, and its root is $s_f$. Indeed,*

$$D = 4A(g_f - g), \qquad g_f - (a + c) = \frac{(a-A)^2}{4A} \ge 0. \tag{36}$$

*The root is positive for $a > A$, negative for $0 < a < A$, and zero for $a = A$. It always lies in $(-1/2, \infty)$ and hence in the DM domain. It is a nonzero double equilibrium only when $a \ne A$.*

*(iii) The two parameter loci meet exactly at the* algebraic degeneracy $a = A$, $g = a + c$. *There* $Q(s) = As^2$ *and*

$$f(s) = -\frac{As^3}{1+s}. \tag{37}$$

*The origin is the sole synchronous equilibrium in the domain at that parameter tuple; it is a triple zero of* $f$ *and a double root of* $Q$*, not an additional nonzero equilibrium.*

*Proof.* The nonzero equation can be written, without approximation, as

$$\mu = As^2 - bs, \qquad g = a + c + bs - As^2 = g_f - A(s - s_f)^2. \tag{38}$$

This is its algebraic continuation, to be filtered by $s > -1$, $g > 0$, and $s \neq 0$ when counting nonzero model equilibria. At $s = 0$ it gives exactly $g = a + c$. Completing the square or using (29) gives $D = 0$, $g = g_f$, and $s = s_f$ as the unique repeated-root condition. The difference of the two parameter values in (36) vanishes exactly at $a = A$. Since $a > 0$, $s_f = a/(2A) - 1/2 > -1/2$. At the common locus, substitution into (25) proves (37) and all the remaining assertions. □

For additional local algebra at the origin, fix $a, A, c$ and vary $g$ (equivalently $\mu$). If $b \neq 0$, the root that tends to zero is

$$s_0(\mu) = \frac{b - \operatorname{sgn}(b)\sqrt{b^2 + 4A\mu}}{2A} = -\frac{\mu}{b} + \frac{A\mu^2}{b^3} + O(\mu^3). \tag{39}$$

This follows by expansion of the explicit square root near $\mu = 0$; $g = a + c - \mu$ remains positive in a sufficiently small neighborhood. Thus nonzero domain-admissible roots approach the origin from the neighboring parameter regions. At the intersection itself,

$$f(s)\big|_{\mu=0} = \frac{s^2(b - As)}{1+s} = bs^2 - as^3 + O(s^4). \tag{40}$$

The coefficient of the usual quadratic scalar term is $b = a - A$. It vanishes when $a = A$; this is precisely the degeneracy above.

More explicitly, for $a = A$ the nonzero polynomial equation is $As^2 = \mu$. For $\mu > 0$ its roots are

$$s_\pm = \pm\sqrt{\mu/A}. \tag{41}$$

The negative one is admissible exactly when $0 < \mu < A$ (equivalently $c < g < a + c$); it is $-1$ when $\mu = A$, and below $-1$ when $\mu > A$. At $\mu = 0$ both labels give the origin; for $\mu < 0$ no real nonzero roots exist. These are algebraic branch descriptions only. Any bifurcation theorem requires the separate work assigned to P09 and P10; none is asserted here.

### 6.7 Canonical parameter corollary

**Corollary 6.6** (Positive branches in the original canonical parameters)**.** *Let* $a, k, c, g, h > 0$, $A = k + h$*, and let* $g_f$ *be as in* (35)*. If* $a \leq k + h$*, there is one simple positive synchronous equilibrium exactly for* $0 < g < a + c$*, and none for* $g \geq a + c$*. If* $a > k + h$*, the exact alternatives are*

| *Parameter range* | *Positive synchronous equilibria* |
|---|---|
| $0 < g \leq a + c$ | *One simple equilibrium,* $s_+\mathbf{1}$ |
| $a + c < g < g_f$ | *Two distinct simple equilibria,* $s_-\mathbf{1}, s_+\mathbf{1}$ |
| $g = g_f$ | *One double equilibrium,* $s_f\mathbf{1}$ |
| $g > g_f$ | *None* |

*In particular, $a+c<g$ does not by itself preclude positive synchrony: two distinct positive equilibria exist if and only if*

$$a > k+h, \qquad a+c < g < a+c+\frac{(a-k-h)^2}{4(k+h)}. \tag{42}$$

*The upper equality gives one repeated positive equilibrium when $a > k+h$; the lower equality gives one simple positive equilibrium and the origin. The origin exists in every row, independently of the displayed positive count. There is no dependence on $n$, $q$, $\gcd(n,q)$, or distinctness of the offsets.*

*Proof.* The sign of $b$ is the sign of $a-(k+h)$; the sign of $\mu$ is the sign of $a+c-g$; and $D = 4A(g_f-g)$ with $A>0$. Substitute these identities into Theorem 6.4. □

### 6.8 Weaker assumptions and limits of the classification

The synchronous substitution and the scalar identity require only homogeneity, finite real coefficients, and nonvanishing denominators. They therefore apply under P01's weaker assumptions as well. For $A>0$, the polynomial formula, discriminant, Vieta relations, and the positive-root alternatives of Theorem 6.4 remain valid for real $a, c, g$: their positive-root proofs use only $A>0$. This observation is not a generalization of the canonical negative-root table. In particular, the domain test (33) uses $a+A>0$, and the bound $s_f > -1/2$ uses $a>0$. P01 allows real $a$ and does not supply these inequalities. The complete domain-filtered classification in this module is intentionally restricted to the canonical positive parameter regime. No classification for $A=0$ or $A<0$ is claimed.

### 6.9 Checks and possible failure points

**Root substitution and discriminant reconstruction.** For $D \geq 0$, put $d=\sqrt{D}$ and $r = (b+\varepsilon d)/(2A)$, $\varepsilon \in \{-1,1\}$. Direct expansion gives

$$\mu + br - Ar^2 = \mu + \frac{2b(b+\varepsilon d)-(b+\varepsilon d)^2}{4A} = \mu + \frac{b^2-d^2}{4A} = 0.$$

For $r \neq -1$, substitution in the original scalar expression gives

$$ar - kr^2 + \frac{cr}{1+r} - \frac{gr}{1+r} - hr^2 = \frac{r}{1+r}(\mu + br - Ar^2) = 0.$$

Only $r > -1$ is retained as a DM equilibrium. If $r=-1$, the two original rational terms are undefined even when $c=g$, so no substitution or canceled extension admits that point. At $\mu=0$, the specialized root $r=b/A$ also gives $br-Ar^2=0$. At $D=0$, $r=s_f=b/(2A)$ gives $\mu+br-Ar^2 = \mu+b^2/(4A)=0$. At $g=c$, the retained root $a/A$ gives $ar-Ar^2=0$. At $b=0$, the square-root branches give $\mu-Ar^2=0$. Finally, direct substitution at $r=0$ gives zero for every parameter tuple. These checks include every exact root formula displayed above, with $s_0$ in (39) being one of $s_\pm$; its truncated expansion is not used as a root.

Independently expanding the polynomial discriminant gives

$$(A-a)^2 - 4A[-(a+c-g)] = a^2+2aA+A^2+4A(c-g) = (a+A)^2+4A(c-g).$$

Also the sum of the two explicit roots is $b/A$, and their product is $(b^2-D)/(4A^2) = -\mu/A$, confirming both Vieta relations.

**Exact representative cells and equality boundaries.** The following rational tuples realize every row of the classification. Set $k = h = 1/2$, hence $A = 1$, and $c = 1$ throughout. All original coefficients are strictly positive. Roots are obtained by the formulas already proved; the counts refer to the preceding table.

| Case | $a$ | $g$ | $\mu$ | $D$ | $(r, d, N_0, N_+)$ |
|---|---|---|---|---|---|
| I | 2 | 1/2 | 5/2 | 11 | $(2, 1, 2, 1)$ |
| II | 2 | 1 | 2 | 9 | $(2, 1, 2, 1)$ |
| III | 2 | 2 | 1 | 5 | $(2, 2, 2, 1)$ |
| IV | 2 | 3 | 0 | 1 | $(1, 1, 2, 1)$ |
| V | 1 | 2 | 0 | 0 | $(0, 0, 1, 0)$ |
| VI | 1/2 | 3/2 | 0 | 1/4 | $(1, 1, 1, 0)$ |
| VII | 1 | 3 | $-1$ | $-4$ | $(0, 0, 1, 0)$ |
| VIII | 2 | 25/8 | $-1/8$ | 1/2 | $(2, 2, 3, 2)$ |
| IX | 2 | 13/4 | $-1/4$ | 0 | $(1, 1, 2, 1)$ |
| X | 1/2 | 49/32 | $-1/32$ | 1/8 | $(2, 2, 1, 0)$ |
| XI | 1/2 | 25/16 | $-1/16$ | 0 | $(1, 1, 1, 0)$ |

Cases II, IV–VI, IX and XI test all relevant equality surfaces, including the simultaneous equality $a = A$, $g = a + c$. For $b < 0, 0, > 0$ inside the $\mu > 0$ cases, one may respectively take $a = 1/2, 1, 2$, and $g = 1/2, 1, 1 + a/2$ for I, II, III. For all three signs of $b$ in VII, take the same three $a$ values and $g = g_f + 1$. These choices also check that omitted sign subcases do not alter the counts. Internal exact-arithmetic checks supplement these displayed examples; they are not numerical dependencies or independent verification of the module.

For reproducibility of that finite check, take $V = \{1/4, 1/2, 1, 2, 3, 4\}$ and all $(a, A, c, g) \in V^4$. For each $(a, A, c) \in V^3$, also take $g \in \{c, a + c, g_f\} + \{-\eta, 0, \eta\}$, where $\eta = \min\{a, A, c, 1\}/64$. The union contains 2,868 distinct positive tuples. Set $k = A/3$, $h = 2A/3$ for this grid; use the displayed $k = h = 1/2$ for the eleven table examples. Exact arithmetic in $u + v\sqrt{D}$, with rational $u, v, D$, checks root residuals, Vieta relations, domain comparisons, and every positive-count equivalence and its converse. Including the eleven table examples, 4,438 original scalar residuals vanish exactly and 217 occurrences of $s = -1$ are explicitly excluded. All eleven cells and all equality surfaces are represented. These finite checks found no counterexample; they supplement rather than establish the exhaustive proof.

**Boundary continuity.** The ordered formulas $s_\pm$ are continuous on the parameter set $D \geq 0$, since $A > 0$. At $D \downarrow 0$ they both tend to $s_f$; for $D < 0$ their absence as real roots is recorded, not bridged by a real continuation. At $\mu \to 0$ with fixed $b > 0$, $(s_-, s_+) \to (0, b/A)$; with fixed $b < 0$, $(s_-, s_+) \to (b/A, 0)$. With $b = 0$ and $\mu \downarrow 0$, both labels tend to zero as in (41). At $g \to c$, $(s_-, s_+) \to (-1, a/A)$; the first root is excluded from the DM domain at the boundary. At $b \to 0$ all limits within $D \geq 0$ are again given by the same continuous formulas. The sign-specific table is therefore compatible with every boundary, although admissible root counts can change there.

**Logical completeness and attempted counterexamples.** The origin was retained before division, and every other equilibrium is equivalent to a nonzero root of $Q$ in $s > -1$. This proves that none was lost or introduced. In particular, the excluded root $-1$ at $g = c$ and the zero root at $\mu = 0$ cannot be counted as additional nonzero equilibria. Positivity, discriminant sign, and root order partition all parameter tuples as in the theorem; each implication in the necessary-and-sufficient statements is checked in every cell. The independent positive-half-line argument supplies a second check. Case VIII is an explicit canonical counterexample to the rejected claim that $a + c \leq g$ rules out positive synchrony. Case IV is a counterexample even at equality. Case X shows that two admissible nonzero roots need not be positive; case XI shows that a repeated domain root need not be positive. Case V prevents counting a repeated zero as a positive equilibrium, and case II prevents admitting a canceled denominator root. All these distinctions are essential.

**Priority checks for the independent P02 auditor.** Independently reconstruct the following steps.

(1) Componentwise substitution with coincident offsets and uniqueness on the maximal interval in $\mathcal{U}$.

(2) The origin before division and equivalence to $Q = 0$ only for $s \neq 0$, $s > -1$.

(3) All discriminant identities, root formulas, and Vieta relations.

(4) The smaller-root domain test at $g = c$, including exclusion of $s = -1$ even when the synchronous rational terms cancel.

(5) Every cell, count, and multiplicity in Theorem 6.3, including $\mu < 0$ and negative roots.

(6) Both directions of every alternative in Theorem 6.4 and its original-parameter corollary.

(7) The branch graph, $g_f$, $s_f$, and the simultaneous degeneracy, distinguishing multiplicity of $Q$ from that of $f$.

(8) All boundary limits and the quadratic coefficient in (40).

(9) The exact representative tuples and original-equation residuals, including the excluded-root and double-counting tests.

(10) The scope of weaker assumptions and the absence of any stability or bifurcation conclusion.

Internal checking and compilation do not change the DRAFT status.

# 7 P03: Origin and homogeneous linearization

## 7.1 TODO: Jacobian at the origin

## 7.2 TODO: Origin stability and the nonhyperbolic boundary

## 7.3 TODO: Homogeneous Jacobian entries and cyclic-shift representation

# 8 P04: Arbitrary-(n,q) Fourier spectrum and normality

## 8.1 TODO: Full complex Fourier spectrum and multiplicities

## 8.2 TODO: Normality of the homogeneous linearization

## 8.3 TODO: Linear stability criterion and its precise nonlinear implications

## 8.4 TODO: Step-q cycle interpretation and index-collision bookkeeping

# 9 P05: Exact step-two mode selection

## 9.1 TODO: Quadratic dispersion representation

## 9.2 TODO: Exact discrete mode-selection rule

## 9.3 TODO: Mode ties, parity cases, and selection along a parameter path

# 10 P06: Large-n selected-wave-number limit

## 10.1 TODO: Selected-wave-number limit

## 10.2 TODO: Discretization error and uniformity domain

# 11 P07: Five-node golden-ratio specialization

## 11.1 TODO: Five-node spectrum and golden-ratio expressions

## 11.2 TODO: Comparison of all five-node mode classes

# 12 P08: Matched bilinear analysis and exact comparison

## 12.1 TODO: MB local well-posedness and orthant invariance

## 12.2 TODO: MB synchronous equilibria and exceptional parameter cases

## 12.3 TODO: Sufficient boundedness and synchronous growth or blow-up regimes

## 12.4 TODO: Full MB Fourier spectrum and step-two mode selection

## 12.5 TODO: Exact theorem-based DM–MB comparison

## 12.6 TODO: Optional archived six-term application gate

## 12.7 TODO: Optional MB-specific Hopf comparison gate

# 13 P09: Transcritical bifurcation at the origin

## 13.1 TODO: Simple synchronous critical direction

## 13.2 TODO: Generic transcritical bifurcation and parameter orientation

## 13.3 TODO: Degenerate steady-state case and scope gate

# 14 P10: Saddle-node bifurcation and true bistable region

## 14.1 TODO: Positive synchronous fold and full-system nondegeneracy

## 14.2 TODO: Full-spectrum branch stability

## 14.3 TODO: Actual bistable parameter set

## 14.4 TODO: Basin-separation claim and its scope

# 15 P11: Hopf spectral conditions and spectral simplicity

## 15.1 TODO: Hopf candidate and nonzero frequency

## 15.2 TODO: Spectral simplicity and exclusion of additional critical modes

## 15.3 TODO: Hypothesis ledger for the eventual Hopf theorem

# 16 P12: Hopf transversality

## 16.1 TODO: Equilibrium derivative along the chosen path

## 16.2 TODO: Total spectral crossing derivative

## 16.3 TODO: Nonzero transversality on the stated parameter set

# 17 P13: First Lyapunov coefficient and nondegenerate Hopf theorem

## 17.1 TODO: Bilinear and trilinear derivative tensors

## 17.2 TODO: Fourier selection rules and resolvent domains

## 17.3 TODO: First Lyapunov coefficient and rigorous sign or nonvanishing

## 17.4 TODO: Assembled nondegenerate Hopf theorem and branch stability conditions

# 18 P14: Equivariant rotating waves

## 18.1 TODO: Exact cyclic spatiotemporal symmetry of the periodic branch

## 18.2 TODO: Isotropy, phase clusters and propagation direction

## 18.3 TODO: Relation of the exact wave symmetry to its leading-order expression

# 19 P15: Persistence under parameter heterogeneity

## 19.1 TODO: Nearby equilibrium existence, uniqueness and smooth dependence

## 19.2 TODO: Positivity and stability persistence with distinct hypotheses

## 19.3 TODO: Optional periodic-orbit persistence gate

# 20 P16: Static sensitivity and localized Green function

## 20.1 TODO: First-order parameter forcing and equilibrium correction

## 20.2 TODO: Fourier sensitivity and static norm bound

## 20.3 TODO: Localized Green function and directional response

## 20.4 TODO: Remainder estimate and its domain

# 21 P17: Transient and harmonic response

## 21.1 TODO: Transient Fourier kernel and Euclidean norm behavior

## 21.2 TODO: Conditional late-time modal dominance

## 21.3 TODO: Harmonic transfer function and frequency response

## 21.4 TODO: Scope of comparison with nonlinear dynamics

# 22 P18: Stability radius and Hopf-threshold sensitivity

## 22.1 TODO: Explicit Jacobian perturbation bound including equilibrium motion

## 22.2 TODO: Sufficient stability radius with controlled remainder

## 22.3 TODO: First-order eigenvalue and Hopf-threshold shift

## 22.4 TODO: Persistence of the relevant Hopf hypotheses under the perturbation

# A N01: Independent validation protocol

Every check below is a future task. Before execution, record the model version, claim IDs, formula version, verified proof dependencies and acceptance criteria. Choose scale-aware error measures and tolerances in advance; do not invent a pass threshold after seeing a result. Numerical agreement does not prove a theorem.

V1. **Analytic Jacobian versus finite differences.** Implement the vector field independently of the analytic Jacobian. Use several perturbation steps and scales, interiors and allowed boundaries, and test coincident offsets at $q = 1$, $q = n - 1$ and even $n$ with $q = n/2$. Gate: P03; P08 for MB.

V2. **Equilibrium residuals.** Record both absolute and scaled residuals, branch labels and admissibility. Test origin, all synchronous branches and near-fold cases; later test heterogeneous equilibria. Gates: P02, P08, P15.

V3. **Direct eigendecomposition versus Fourier eigenvalues.** Match unordered spectra with multiplicity; check real and imaginary parts, conjugacy, even-$n$ singleton modes, ties, and $\gcd(n, q) > 1$. Gate: P04; P05–P08 for selection and comparison.

V4. **Symmetry/equivariance tests.** Test the homogeneous vector field and Jacobian against the fixed group action and the synchronous restriction; later test exact wave phase relations. Gates: P02–P04 and P14.

V5. **Continuation versus direct simulation.** Compare stable equilibrium and periodic branches, periods and amplitudes. For unstable branches use residuals and variational/Floquet checks; lack of convergence in forward simulation is not a branch validation. Gates: P09–P14 as applicable.

V6. **First Lyapunov coefficient from two independent implementations.** Use independently assembled tensors or derivative methods and a separate normal-form implementation. Record all inner-product, scaling, frequency, factorial and software conventions. Shared code paths do not count as independence. Reconcile magnitudes as well as signs. Gate: P13.

V7. **Convergence under tighter tolerances.** Refine integration steps, relative/absolute tolerances, continuation mesh, arithmetic precision, finite difference increments and sampling horizons as appropriate. Record convergence of each reported observable; a default tolerance is not evidence.

V8. **Reproducibility and software-version recording.** Archive exact versions, dependency lockfiles, operating system, precision, solver choices, seeds, parameters, initial data, time horizons, continuation controls, raw outputs, figure scripts and code/version hashes. A fresh environment must reproduce outputs within the predeclared tolerance.

V9. **Additional theorem-specific checks.** Boundary-face and box tests after P01; tensor derivatives and transversality after P12–P13; Floquet multipliers and phase shifts after P14; first-order remainder rates and Green kernels after P15–P16; transient and harmonic responses after P17; stability bounds and threshold shifts after P18. Record tests near failure modes.

### A.1 Per-check result form to fill later

Claim and theorem version; independent implementation identifiers; inputs; observable and norm; predicted result; predeclared tolerance; measured error; refinement table; pass/fail/unresolved; failure diagnosis; reviewer and date. No field contains a computed result at initialization.

## B N02: Continuation plan

1. Resolve B02, B06 and NB11–NB12: choose branch, parameter path, admissible parameter range, software, precision, event definitions and stopping criteria. The master baseline and Hopf coordinates are unverified seed suggestions; retrieve their exact archival values from the registry.

2. After P02–P04, continue all admissible synchronous equilibrium branches and retain the origin. Tag residuals, full spectra, mode labels and every inadmissible segment. Use P08 for a separately implemented MB comparison.

3. After P09–P10, identify transcritical and fold candidates and compare the continuation with proved statements. Separate root multiplicity, full stability and any basin claim. A sampled basin picture cannot establish a global separatrix or exclude additional attractors.

4. After P11–P13 and N01 validation, continue a periodic branch from the verified Hopf setting. Record amplitude convention, period, mesh convergence, Floquet multipliers, neutral phase multiplier and transverse stability.

5. After P14, check the exact cyclic time-shift relation with the declared orientation, phase convention and possible phase clusters.

6. For two-parameter work, choose the plane before plotting. Continue only curves justified by the completed analysis. Label fold–Hopf, zero–Hopf, Hopf–Hopf, generalized-Hopf and mode-tie points as candidates until separately validated; no existence of such points is assumed in this plan.

7. After P15–P18, continue selected heterogeneous equilibria and Hopf thresholds for explicitly named profiles. Compare first-order predictions and remainders across decreasing amplitudes, keeping coefficients positive.

### B.1 Required continuation record

Model and proof versions; full parameters; network size and step; branch and seed provenance; free parameters; step limits; mesh and collocation settings; solver tolerances and precision; event rules; raw branch files; failed runs; independent checks and exact scripts to regenerate the diagrams.

## C N03: Figure plan

Every caption must identify what is theorem, computation or interpretation. Use a figure only after its stated dependencies and N01 checks are satisfied.

| ID | Planned figure | Evidence gate and required disclosure |
|---|---|---|
| F1 | Network family | Canonical offsets and indexing; mark the connected-step condition, offset coincidences and $(5, 2)$ specialization. No dynamical claim. |
| F2 | Interaction laws | Canonical definitions and B05; declare fixed source/receiver values and calibration. Do not label the input standard Holling type II. |
| F3 | Synchronous branches | P02, P03, P09, P10; distinguish existence, stability, two-root region and any proved bistability. |
| F4 | Dispersion and selection | P04–P07; all mode classes, ties, even-$n$ cases, large-$n$ family and exact error quantity. |
| F5 | DM–MB comparison | P01–P05, P08; matched parameter plane, eligibility of equilibria and full-spectrum stability. |
| F6 | Two-parameter organization | P09–P13; separate proved curves, numerical continuation and unresolved special points. |
| F7 | Hopf and rotating waves | P11–P14; amplitude, period, Floquet multipliers, trajectories, phase-shift errors and tolerance refinement. |
| F8 | Heterogeneity and response | P15–P18; named perturbation profiles, static Green function, transient/harmonic response, first-order errors and stability/threshold bounds. |
| F9 | Five-node synthesis | P07, P08 and the applicable verified bifurcation/response modules; no physiological inference from algebraic constants. |

### C.1 Figure record to fill later

Claim IDs; verified theorem versions; model, parameters and calibration; data provenance; initial data; solver and continuation settings; tolerances; random seed if any; code version; reproduction command; error bars or error metrics; accessible labels and caption; independent checker. Optional random-prior maps and extra network sizes remain supplementary choices. No favorable-regime-only selection is prescribed.

## D Adversarial mathematical audit template

**Audit ID, reviewer, date and independence:**

Record whether the reviewer is independent of the derivation and which version or hash is reviewed.

**Statement being audited:**

Exact theorem text, claim registry IDs, proof-file path and statement label. Do not audit a paraphrase.

**Exact assumptions:**

Every quantified domain, parameter restriction, regularity hypothesis, branch, norm, symmetry, exclusion and nondegeneracy.

**Dependency verification:**
List each input theorem, exact version, VERIFIED/AUDITED status and applicability; identify circular reasoning.

**Line-by-line proof checks:**
For each step record justification, algebra, signs, quantifiers, logical implication, and missing hypotheses. Require independent reconstruction of critical calculations.

**Edge cases:**
Zero and repeated roots; parameter boundaries; even/odd n; q=1, q=n-1 and q=n/2 when applicable; gcd cases; coincident shifts; self-conjugate modes; ties and resonances; singular denominators; state boundary.

**Counterexample search:**
Challenge the strongest quantifier; try excluded and limiting cases; distinguish a successful proof from failure to find a counterexample. Record whether the search was analytic or computational.

**Symbolic checks:**
Independent derivatives, identities, normalization, adjoints and remainders; source of any computer-algebra expression.

**Numerical checks:**
Independent implementation, residuals, conditioning, precision and convergence; whether ordinary or rigorously validated numerics; which analytic assertion the check can and cannot support.

**Literature overlap:**
Exact source and location, hypotheses, model dictionary, novelty limitations and any result imported by citation.

**Verdict:**
Select exactly one: **verified** / **revision needed** / **false** / **unresolved**. Explain the decisive evidence and remaining obligations.

**Disposition and sign-off:**
Corrections, owner, downstream invalidations, recheck version and date; do not mark AUDITED until the verdict is verified.